METALLURGY OF COPPER

Exam-Pattern Answers

Q.1 Extraction of Copper from Sulphide Ore

In the extraction of copper from its sulphide ore, the metal is finally obtained by reduction of cuprous oxide. [2 + 2 + 1]

a. Which process is used to convert sulphide ore into the oxide? Explain and write the reactions.

The process used is ROASTING. The concentrated ore (obtained after froth floatation) is heated strongly in a reverberatory furnace in excess of air. Roasting removes moisture, sulphur, arsenic and antimony as their volatile oxides, and partially converts the sulphide ore into oxide.

$$2CuFeS_2 + O_2 \rightarrow Cu_2S + 2FeS + SO_2$$

$$2Cu_2S + 3O_2 \rightarrow 2Cu_2O + 2SO_2$$

$$2FeS + 3O_2 \rightarrow 2FeO + 2SO_2$$

b. Action of blister copper on (i) air at high temperature and (ii) dilute HNO_3.

(i) With air at high temperature: The surface oxidises to Cu_2O; the dissolved SO_2 escapes violently through the molten metal producing blister-like swellings on the surface — hence the name 'blister copper'.

(ii) With dilute nitric acid: Copper dissolves, liberating nitric oxide gas.

$$3Cu + 8HNO_3(\text{dilute}) \rightarrow 3Cu(NO_3)_2 + 2NO\uparrow + 4H_2O$$

c. Equation for the reaction of Cu_2S with the final blast of air.

$$2Cu_2S + 3O_2 \rightarrow 2Cu_2O + 2SO_2$$

$$2Cu_2O + Cu_2S \rightarrow 6Cu + SO_2\uparrow \quad \text{(self-reduction / auto-reduction step)}$$

Q.2 Extraction of Copper from Copper Pyrites

The ore is concentrated by froth floatation, then heated in a furnace; Cu_2S is converted into copper using a blast of air. [2 + 1 + 2]

a. How do iron impurities separate from the sulphide ore?

After roasting, the roasted ore (Cu_2S + FeS + FeO + gangue SiO_2) is mixed with coke and silica (flux) and smelted in a blast/smelting furnace. FeO combines with silica to form fusible slag (iron silicate) which, being lighter, floats above and is skimmed off, leaving behind molten Cu_2S and FeS as 'copper matte'.

$$FeO + SiO_2 \rightarrow FeSiO_3 \text{ (slag)}$$

b. Why is complete removal of FeS necessary?

If FeS is not completely removed at the smelting stage, it would also oxidise during Bessemerisation along with Cu_2S, and the resulting iron oxide/silicate would contaminate the blister copper, lowering its purity. Complete removal of FeS ensures that the matte fed to the converter is essentially pure Cu_2S, giving cleaner self-reduction to metallic copper.

c. Obtaining blister copper from copper matte (Bessemerisation).

Molten copper matte is charged into a Bessemer converter (a pear-shaped, tiltable, basic-lined vessel fitted with tuyeres). A blast of hot air mixed with sand is blown through the molten matte.

$$2Cu_2S + 3O_2 \rightarrow 2Cu_2O + 2SO_2$$

$$2Cu_2O + Cu_2S \rightarrow 6Cu + SO_2\uparrow \quad \text{(blister copper, \textasciitilde 98\% pure)}$$

[Figure: Pear-shaped Bessemer converter, mounted on trunnions so it can tilt; tuyeres at the base admit the air blast into the molten matte; slag floats on top and is poured off, while SO_2 escapes from the mouth.]

Q.3 Metal 'X' of Group 11

Metal 'X' lies in group 11 and is obtained from its ores by metallurgical processes. [2 + 1 + 1 + 1]

a. Identify metal X. Nature of crystals formed with $AgNO_3$ solution; reaction involved.

X = Copper (Cu). When a strip of copper is dipped into silver nitrate solution, copper, being more reactive than silver, displaces silver from the solution. The silver is deposited as shining crystals (metallic silver), while the solution turns blue due to formation of copper nitrate.

$$Cu + 2AgNO_3 \rightarrow Cu(NO_3)_2 + 2Ag\downarrow \text{ (silver crystals)}$$

b. A reaction in which the metal acts as a reducing agent.

$$Cu + 2AgNO_3 \rightarrow Cu(NO_3)_2 + 2Ag$$

Here copper reduces Ag^+ ions to metallic silver (and is itself oxidised to Cu^{2+}), so copper acts as the reducing agent.

c. Action of ammonia solution on the sulphate of metal X ($CuSO_4$).

With a small amount of NH_4OH, a pale-blue precipitate of $Cu(OH)_2$ is formed:

$$CuSO_4 + 2NH_4OH \rightarrow Cu(OH)_2\downarrow + (NH_4)_2SO_4$$

With excess NH_4OH, the precipitate dissolves to give a deep-blue solution of tetraamminecopper(II) sulphate:

$$Cu(OH)_2 + 4NH_4OH \rightarrow [Cu(NH_3)_4](OH)_2 + 4H_2O$$

d. Two uses of metal X (Copper).

- Used for making electrical wires and cables, owing to its high electrical conductivity.
- Used for making alloys such as brass (Cu + Zn) and bronze (Cu + Sn), and for utensils and coins.

Q.4 Electrolytic Purification of Copper

Purification of copper by electrolysis. [3 + 2]

a. Anode and cathode used; main electrode reactions.

Anode: a thick block of impure copper. Cathode: a thin strip of pure copper. Electrolyte: acidified copper sulphate solution.

$$\text{Anode:} \quad Cu \rightarrow Cu^{2+} + 2e^- \quad \text{(impure copper dissolves)}$$

$$\text{Cathode:} \quad Cu^{2+} + 2e^- \rightarrow Cu \quad \text{(pure copper is deposited)}$$

[Figure: Electrolytic cell — impure copper block (anode, +) and pure copper strip (cathode, −) suspended in acidified $CuSO_4$ solution, both connected to a battery; anode mud collects below the anode.]

b. Fate of zinc impurity (above copper in the electrochemical series).

Zinc, being more reactive than copper, dissolves from the anode as Zn^{2+} ions along with copper. However, at the cathode only Cu^{2+} ions are preferentially discharged (reduced) because copper is less reactive; Zn^{2+} ions are not deposited and instead remain dissolved in the electrolyte, whose concentration of zinc sulphate gradually increases. (Less reactive impurities such as silver and gold, being below copper, are not oxidised at all and simply fall to the bottom as 'anode mud'.)

Q.5 Solution 'X' — Copper Sulphate

A light-blue precipitate $Cu(OH)_2$ is obtained by adding caustic soda to solution 'X'. [1 + 1 + 3]

a. Chemical formula of X; commercial method of preparation.

X = $CuSO_4$ (copper sulphate). Commercially it is prepared by roasting copper scrap/ore in air in presence of dilute sulphuric acid (air is blown through a mixture of copper and dilute H_2SO_4):

$$2Cu + 2H_2SO_4 + O_2 \rightarrow 2CuSO_4 + 2H_2O$$

b. Crystallisation of X; an equation showing its acidic nature.

On crystallisation, $CuSO_4$ forms blue crystals of $CuSO_4 \cdot 5H_2O$ (blue vitriol). Its acidic nature is shown by its reaction with a carbonate, which liberates carbon dioxide:

$$2CuSO_4 + 2Na_2CO_3 + H_2O \rightarrow CuCO_3.Cu(OH)_2\downarrow + 2Na_2SO_4 + CO_2\uparrow$$

c. Heating crystallised X at 230°C; reaction with (i) ammonia and (ii) aq. KI.

On heating $CuSO_4 \cdot 5H_2O$ at about 230°C, all the water of crystallisation is lost and white anhydrous $CuSO_4$ is formed:

$$CuSO_4.5H_2O \xrightarrow{\text{heat, } 230°C} CuSO_4 \text{ (white, anhydrous)} + 5H_2O$$

(i) With ammonia (excess): a deep-blue complex is formed —

$$CuSO_4 + 4NH_4OH \rightarrow [Cu(NH_3)_4]SO_4 + 4H_2O$$

(ii) With aqueous KI: a white precipitate of cuprous iodide is formed along with liberation of iodine —

$$2CuSO_4 + 4KI \rightarrow Cu_2I_2\downarrow \text{ (white)} + 2K_2SO_4 + I_2$$

Q.6 Blue Vitriol ($CuSO_4 \cdot 5H_2O$)

Penta-hydrated copper sulphate is called blue vitriol. [1 + 1 + 1 + 2]

a. Preparation of blue vitriol starting from metallic copper.

Copper is dissolved in hot concentrated sulphuric acid; the resulting solution is evaporated and cooled to obtain blue crystals.

$$Cu + 2H_2SO_4 \text{ (conc., hot)} \rightarrow CuSO_4 + SO_2\uparrow + 2H_2O$$

On slow evaporation and cooling of the $CuSO_4$ solution, blue crystals of $CuSO_4 \cdot 5H_2O$ separate out.

b. Conversion of blue vitriol into black oxide.

$$CuSO_4.5H_2O \xrightarrow{\text{heat}} CuSO_4 \xrightarrow{\text{strongly heated}} CuO \text{ (black)} + SO_3\uparrow$$

c. Why hydrated copper sulphate is called 'blue vitriol'.

It is called blue vitriol because it is a deep-blue crystalline hydrated sulphate salt — 'vitriol' being the traditional name given to glassy, vitreous-looking sulphate compounds — and its characteristic blue colour arises from the five molecules of water of crystallisation coordinated around the Cu^{2+} ion.

Q.7 Precipitates from $CuSO_4$ and Caustic Soda

A light-blue precipitate 'A', obtained by adding caustic soda to a solution of cupric sulphate, is converted to a black precipitate 'B' on heating. [1 + 1]

Identify A and B; write the reactions.

A = $Cu(OH)_2$ (cupric hydroxide) — light-blue precipitate.

B = CuO (cupric oxide) — black precipitate.

$$CuSO_4 + 2NaOH \rightarrow Cu(OH)_2\downarrow \text{ (A)} + Na_2SO_4$$

$$Cu(OH)_2 \text{ (A)} \xrightarrow{\text{heat}} CuO \text{ (B)} + H_2O$$

Q.8 Identification of Ore X and Metal M

X is an ore of metal M. Roasting of X gives a black oxide W (Group II basic radical) and metal M with a gas as major byproduct; the gas turns acidified $K_2Cr_2O_7$ solution green. [1 + 1 + 1 + 2]

a. Identify metal X (the ore).

Since the gas turns acidified potassium dichromate solution from orange to green, the gas is SO_2, confirming a sulphide ore. The black oxide W (a Group II basic radical) is CuO.

X = Copper glance, Cu_2S (ore). M = Copper (Cu).

b. Reaction during calcination/roasting of X.

$$2Cu_2S + 3O_2 \rightarrow 2Cu_2O + 2SO_2\uparrow$$

$$2Cu_2O + Cu_2S \rightarrow 6Cu + SO_2\uparrow$$

(Partial oxidation of Cu_2S gives Cu_2O, which is then reduced by the remaining Cu_2S itself to give metallic copper directly, along with SO_2 gas.)

c. Action of the gas (SO_2) on acidified $K_2Cr_2O_7$.

SO_2 reduces the orange dichromate ion to green chromic sulphate:

$$K_2Cr_2O_7 + H_2SO_4 + 3SO_2 \rightarrow K_2SO_4 + Cr_2(SO_4)_3 \text{ (green)} + H_2O$$

d. Converting X into its vitriol.

The copper obtained is dissolved in hot concentrated sulphuric acid and the solution crystallised:

$$Cu + 2H_2SO_4 \text{ (conc.)} \rightarrow CuSO_4 + SO_2\uparrow + 2H_2O$$

On cooling and crystallisation, blue vitriol ($CuSO_4 \cdot 5H_2O$) is obtained.

Q.9 Coinage Metal A and Ammonia

Ammonia solution added to the sulphate of coinage metal A gives blue precipitate B, which dissolves in excess reagent to form deep-blue solution C. [2 + 2 + 1]

a. Identify A, B, C with the sequence of reactions.

A = Copper (present as $CuSO_4$, cupric sulphate).

B = $Cu(OH)_2$ (pale/light-blue precipitate).

C = $[Cu(NH_3)_4]SO_4$, tetraamminecopper(II) sulphate (deep-blue solution).

Step 1 — small amount of NH_4OH added:

$$CuSO_4 + 2NH_4OH \rightarrow Cu(OH)_2\downarrow \text{ (B)} + (NH_4)_2SO_4$$

Step 2 — excess NH_4OH added, precipitate B dissolves:

$$Cu(OH)_2 \text{ (B)} + (NH_4)_2SO_4 + 2NH_4OH \rightarrow [Cu(NH_3)_4]SO_4 \text{ (C)} + 4H_2O$$

b. Purification of metal A — Electrolytic Refining.

Anode: impure copper block. Cathode: thin strip of pure copper. Electrolyte: acidified $CuSO_4$ solution.

$$\text{Anode: } Cu \rightarrow Cu^{2+} + 2e^-$$

$$\text{Cathode: } Cu^{2+} + 2e^- \rightarrow Cu$$

Pure copper is deposited at the cathode. Less reactive impurities (Ag, Au) fall as anode mud; more reactive impurities (Zn, Fe) remain dissolved in the electrolyte.

c. Action of KI on a solution of the sulphate of metal A.

$$2CuSO_4 + 4KI \rightarrow Cu_2I_2\downarrow \text{ (white)} + 2K_2SO_4 + I_2$$

The liberated iodine imparts a brown colour to the solution; addition of starch gives a blue-black colouration, confirming its presence.